

	1	2	3	4	5	6	7	8	
A	Integrated draft: power chain (sheets 2–4) and board stage (sheets 5–8). See DESIGN_SPEC.md. Inputs: VBAT 6–25.2 V (2–6S) from the MicoAir AM32 ESC pad row J3, and USB VBUS. Rails: 5V_IN from the U26 AP63205 buck, V5_SYS from the U25 TPS2121 mux, V3V3_SYS from the TPS62913 buck, V3V3_ANA from the TPS7A20 LDO. Servo 5 V is 5V_IN, upstream of the mux. No PCB placement or routing has been performed.								A
B	POWER 1 / ESC pad row, VBAT buck, USB, priority power mux File: power_sources.kicad_sch				BOARD 2 / IMU, barometer, high-g File: sensors.kicad_sch				B
C	POWER 2 / 3.3 V system buck and analog LDO File: power_3v3.kicad_sch				BOARD 3 / log flash and microSD File: storage.kicad_sch				C
D	POWER 3 / peripheral ports File: power_periph.kicad_sch				BOARD 4 / servo outputs PWM5–8 File: actuators.kicad_sch				D
E	BOARD 1 / RP2354B File: mcu.kicad_sch								E
F	1	2	3	4	5	6	7	8	

Sheet: /
File: MARV-V2.kicad_sch

Title: MARV integrated power – preliminary

Size: A3

Date: 2026-09-14

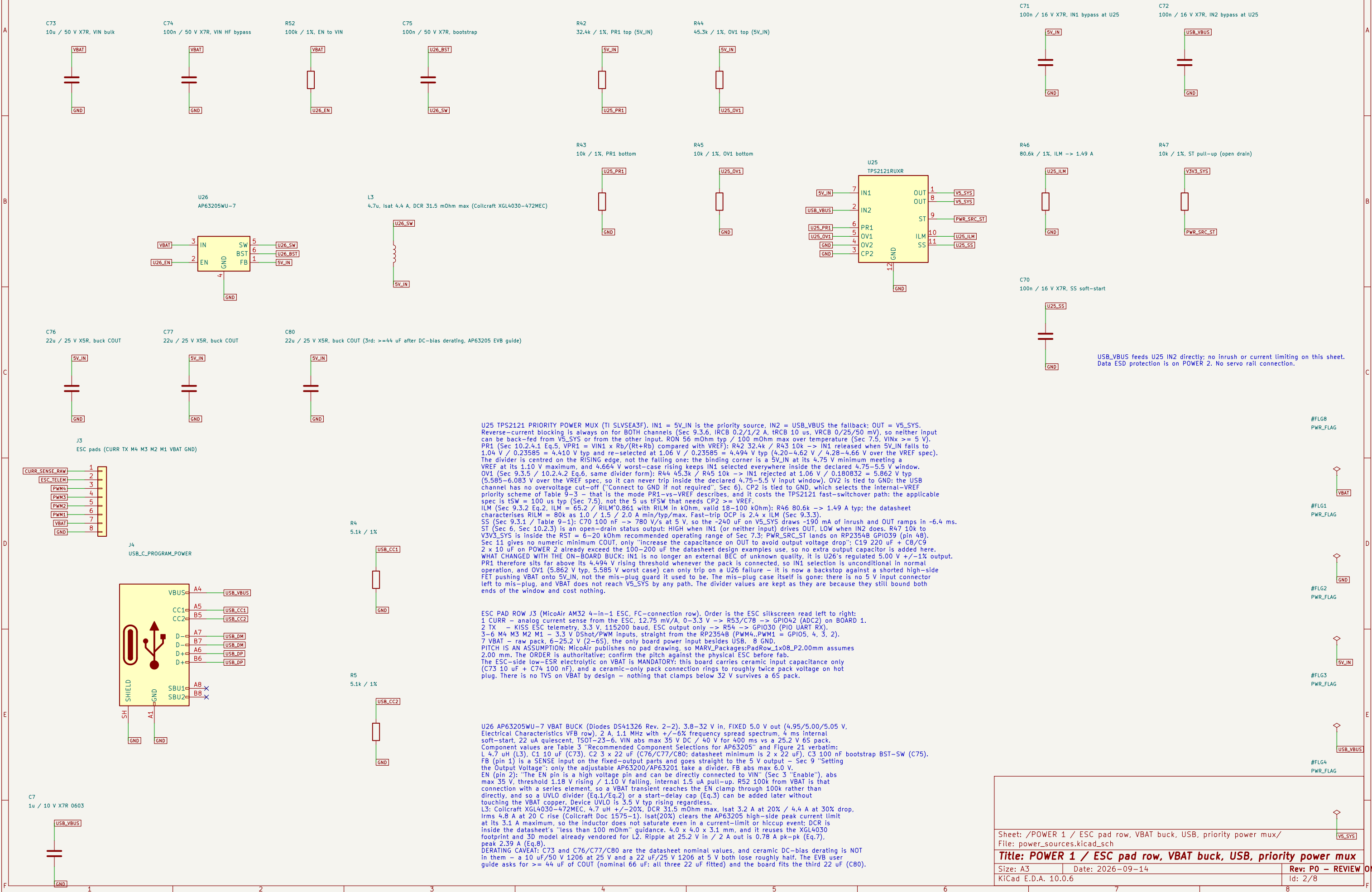
Rev: P0 – NOT FOR

KiCad E.D.A. 10.0.6

Id: 1/8

Sheet: / File: MARV-V2.kicad_sch		
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Size: A3	Date: 2026-09-14	Rev: P0 – NOT FOR FAB
KiCad E.D.A. 10.0.6	Id: 1/8	

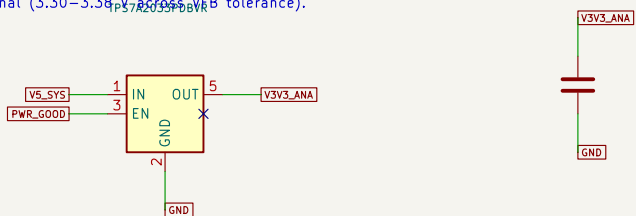
BOARD INPUTS: VBAT 6–25.2 V (2–6S) from the ESC pad row J3, and USB VBUS ~5 V (J4). U26 (AP63205WU) makes 5V_IN from VBAT; 5V_IN is the priority input of the U25 power mux and also the servo 5 V bus.



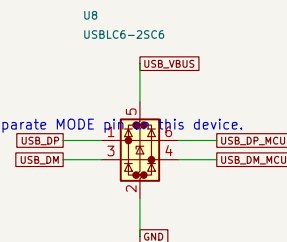
AVIONICS ONLY: 300 mA continuous / 500 mA short peak, provisional. NO SERVO POWER.



V3V3_SYS feedback (TPS62913 datasheet, Sec. B.2.2.2.6, Eq.8): $V_{OUT} = V_{FB} \times (1 + R1/R2)$; $V_{FB} = 0.8 \text{ V typ. } 0.792\text{--}0.812 \text{ V spec. (Sfr. 6.5)}$
 $R2 = 4.99 \text{ k}\Omega \text{ (}\leq 5 \text{ k}\Omega \text{ per datasheet noise guidance). } R1 = R2 \times (V_{OUT}/V_{FB} - 1) = 4.99 \text{ k} \times (3.3/0.8 - 1) = 4.99 \text{ k} \times 3.125 = 15.59 \text{ k}\Omega \rightarrow \text{nearest } 1\% \text{ E96} = 15.8 \text{ k}\Omega \text{.}$
 Actual $V_{OUT} = 0.8 \text{ V} \times (1 + 15.8 \text{ k}/4.99 \text{ k}) = 3.33 \text{ V nominal (}3.30\text{--}3.38 \text{ V per } 1\% \text{ E96 tolerance).}$

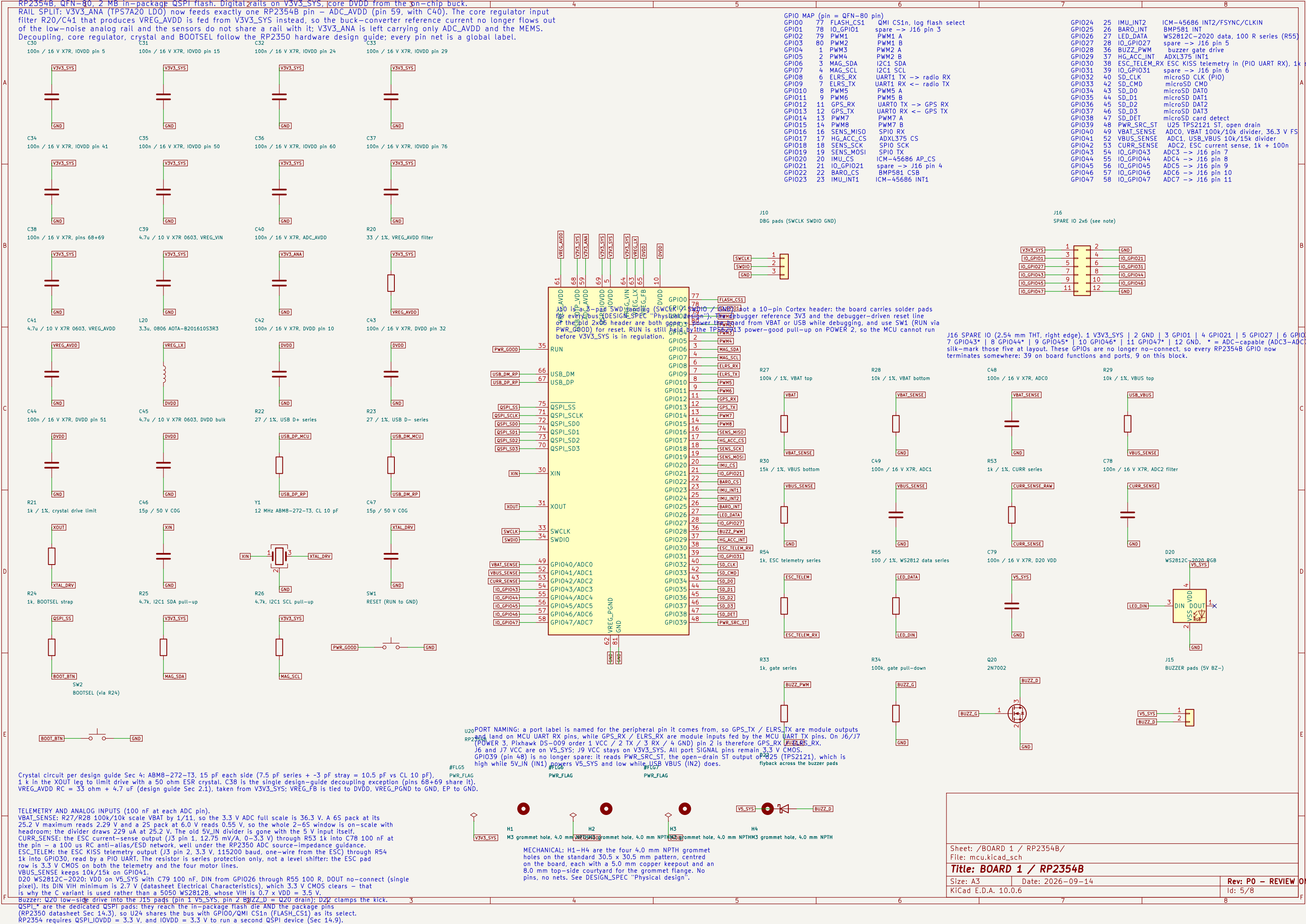


S-CONF = 6.04 kΩm to GND (Table 7-1); 2.2 MHz switching; triangle spread-spectrum ON, output discharge OFF, no external sync.
2.2 MHz + 2.2uH matches the VIN=5V/VOU1<=3.3V design table (Table 8-2). TP562913 runs fixed-frequency PWM at all loads, no light-load skip mode (Sec 7.4.1) -- forced PWM is inherent; there is no separate MODE pin on this device.
V0 (pin 3) senses the node between L1 and the ferrite bead (device internal load); the FB divider senses VV3_VSYS after the bead for low-noise remote regulation (Sec 7.1 / 8.2.2.2.4).

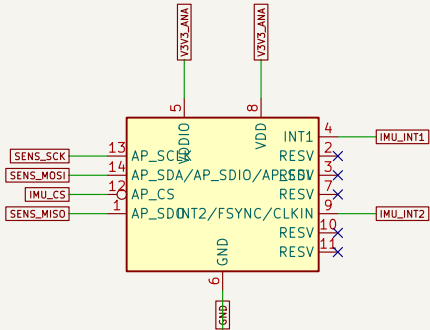


V3V3_ANA feeds ICM-45686, BMP58, ADXL375 and the RP23548 ADC_AVDD pin (100 nF at each pin on the MCU sheet); VREG_AVDD is filtered from V3V3_VSYS instead, so the analog rail carries no core-regulator current. I2C_EN (pin 3) is on PWR_GOOD rather than VSYS; the analog rail therefore starts only once the TPS62913 declares V3V3_VSYS in regulation, and drops with it, which removes the window where the sensors were biased from an unregulated VSYS while the MCU was still held in reset.

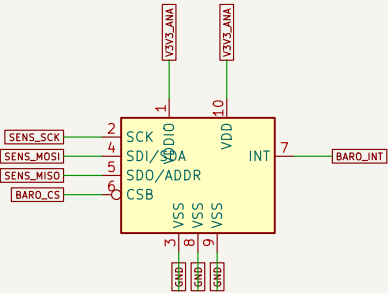




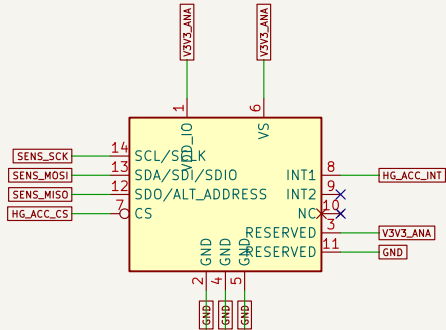
All three sensors run 4-wire SPI on the shared SENS_SCK / SENS_MOSI / SENS_MISO bus with one chip select each, and are powered from V3V3_ANA (TPS7A20 LDO). Every supply pin on this sheet gets 100 nF + 1 uF; BMP581 and ADXL375 because their own datasheets call for the pair, the ICM-45686 as a deliberate addition – its datasheet BOM (DS-000577 Rev 1.0 Table 11) lists 0.1 uF only.



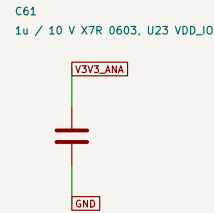
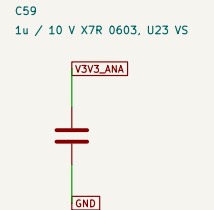
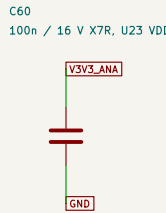
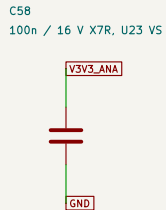
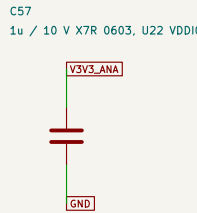
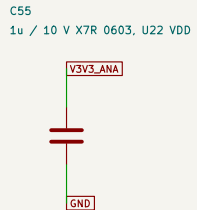
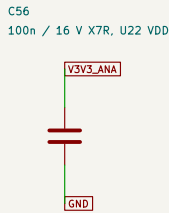
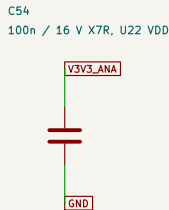
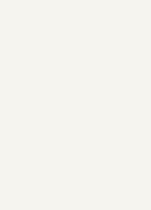
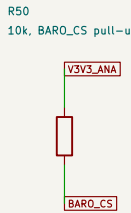
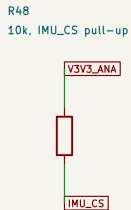
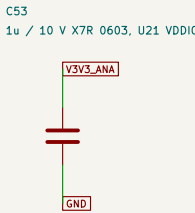
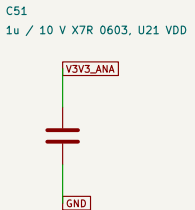
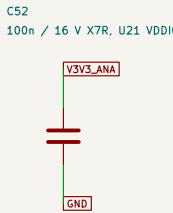
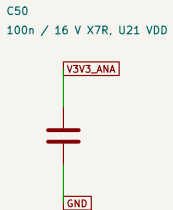
U21
ICM-45686 6-axis IMU



U22
BMP581 barometer



U23
ADXL375 high-g accelerometer



U21 ICM-45686 (TDK InvenSense DS-000577 Rev 1.0, Table 10 "Signal Descriptions" / Figure 4 pin-out, Single Interface SPI mode, Figure 10): AP_CS (pin 12) = IMU_CS, AP_SCLK (13) = SENS_SCK, AP_SDA/AP_SDIO/AP_SDI (14) = SENS_MOSI, AP_SDO/AP_ADO (1) = SENS_MISO, INT1 (4) = IMU_INT1, INT2/FSYNC/CLKIN (9) = IMU_INT2 – typed bidirectional because FSYNC/CLKIN are host-driven inputs while INT2 is a chip output; only the INT2 function is used here, VDD (8) and VDDIO (5) are both on V3V3_ANA; GND (6) is the only ground pin. Pins 2, 3, 7, 10 and 11 (RESV) are all left No Connect; Table 10 allows No Connect / VDDIO / GND for every RESV pin, and the Single-Interface SPI typical application schematic (Figure 10) shows all five left unterminated (internal pull enabled by default, per the per-pin notes in Table 10); they are only needed for the AUX1 or I2C-master dual-interface modes, neither of which this board uses. C50 (VDD) and C52 (VDDIO) are the 100 nF X7R of Table 11 "Bill of Materials" (C1/C2 in the datasheet); C51 (VDD) and C53 (VDDIO) add 1 uF on top of them. The datasheet BOM lists 0.1 uF only; this board keeps 100 nF + 1 uF per supply pin for consistency with U22/U23 and common flight-controller practice.

U22 BMP581: CSB = BARO_CS, SDI = SENS_MOSI, SDO = SENS_MISO, SCK = SENS_SCK, INT = BARO_INT, VDDIO and VDD on V3V3_ANA, all three VSS pins to GND.
U23 ADXL375: CS = HG_ACC_CS, SDA/SDI = SENS_MOSI, SDO = SENS_MISO, SCL/SCLK = SENS_SCK, INT1 = HG_ACC_INT. Per the pin function table, pin 3 (RESERVED) goes to VS and pin 11 (RESERVED) to GND; pin 10 (NC) and the unused INT2 are left open. VS and VDD I/O are both on V3V3_ANA.

R48, R50, R51: 10k pull-ups to V3V3_ANA on the three SPI chip selects (IMU_CS, BARO_CS, HG_ACC_CS). R49 (the former second IMU chip-select pull-up) is removed along with that net – the ICM-45686 has a single chip select. Same rationale as R35 on FLASH_CS1 (BOARD_3): every sensor is held deselected before the RP2354B drives the pin, so a GPIO left floating through reset, BOOTSEL or a debugger halt cannot open a transaction on the shared SENS_SCK / SENS_MOSI / SENS_MISO bus. They bias to V3V3_ANA, the same rail as every VDDIO pin on this sheet, so there is no pull-up current path into an unpowered I/O supply and no CS pin is pulled above its own VDDIO during rail sequencing.

Sheet: /BOARD 2 / IMU, barometer, high-g/
File: sensors.kicad_sch

Title: BOARD 2 / IMU, barometer, high-g

Size: A3
KiCad E.D.A. 10.0.6

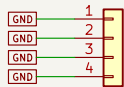
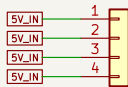
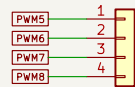
Date: 2026-09-14

Rev: P0 – REVIEW ONLY

Id: 6/8

Sheet: /BOARD 3 / log flash and microSD/			ONLY
File: storage.kicad_sch			
Title: BOARD 3 / log flash and microSD			
Size: A3	Date: 2026-09-14	Rev: P0 - REVIEW	
KiCad E.D.A. 10.0.6		Id: 7/8	

PWM5-8 only. PWM1-4 leave the board on the ESC pad row (J5, POWER 1) and appear nowhere on this sheet.	4	5	6	7	8
Three 1x04 rows on 2.54 mm centres form the usual servo block: signal / 5 V / GND. No series resistors – 3.3 V logic straight from the RP2354B, which is what servos and ESCs expect.					



Row 2 is the 5V_IN net – the output of U26 (the AP63205 VBAT buck on POWER_1), taken BEFORE U25, the TPS2121 priority power mux. Servo current therefore never crosses V5_SYS or the mux and is not counted against the mux ILM current limit (1.49 A, set by R46); it does count against the buck, which is a 2 A part, so the four servo channels share whatever is left of 2 A after the avionics load (~0.3 A typical, 0.6 A peak). Sizing the 5V_IN copper and budgeting servo current against U26 is an open item in DESIGN_SPEC.
Nothing on this sheet loads V3V3_SYS or USB_VBUS.

Three stacked Conn_01x04 symbols (J12 signal, J13 5 V, J14 GND) with three 1x04 THT footprints: no 3-row generic connector symbol or 2.54 mm 3x04 footprint ships with this KiCad install. Place them on 2.54 mm centres at the bottom-left of the board to form a 3x4 servo block in standard servo lead order (SIG / 5 V / GND); column n is J12 pin n / J13 pin n / J14 pin n. Through-hole 2.54 mm is deliberate here: DESIGN_SPEC "Physical design" puts solder pads on every bus and keeps THT pins for leftover IO only.